Lee Zhi Cheng

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Education

Tsinghua University Sep 2023 – Jun 2025
Master's in Advanced Computing, Department of Computer Science and Technology
GPA 3.80/4.00 | Chinese Government Scholarship | KEG Lab, advised by Prof. Juanzi Li

Shanghai Jiao Tong University Sep 2017 – Jun 2021
Bachelor of Engineering in Automation
GPA 3.44/4.30 (83.73/100) | Shanghai Municipal Scholarship, Third Prize

Technical Expertise

- **Agentic RAG.** Experience in tool-use trajectory generation, supervised fine-tuning, and model evaluation. Fine-tuned GLM-4-9B for iterative retrieval and multi-step reasoning. Co-first author of ReaRAG (IJCNN 2026) and DiaKoP (CIKM Demo 2024).
- **Agent Harness.** Multi-agent orchestration, context isolation, MCP tool integration, and execution feedback loops using the Claude Agent SDK and Pi.
- **Evaluation.** Multi-hop question answering, LLM-as-a-Judge evaluation, and failure analysis using retrieval traces and tool outputs.

Experience

Zhipu AI (Z.AI) | Algorithm Engineer, LLM Agents Aug 2025 – Present

- Led the design and development of enterprise agent systems for VC research and automotive unit-test generation, focusing on agent harness design, multi-agent orchestration, and execution feedback loops.

Zhipu AI (Z.AI) | LLM Research Intern Jul 2024 – May 2025

- Conducted agentic RAG research on multi-step reasoning, tool use, and retrieval augmentation; developed training data, performed SFT, and evaluated model performance. The work resulted in ReaRAG, published at IJCNN 2026 (co-first author).

Wise AI Malaysia | AI Engineer Nov 2021 – Aug 2023

- Led data preparation, training, and evaluation for a ResNet face anti-spoofing model that passed third-party Level 1 presentation attack testing under ISO/IEC 30107.

Research Experience

ReaRAG | Agentic RAG for Multi-Hop Reasoning Jul 2024 – May 2025
Co-first author, IJCNN 2026

- **Agent trajectories.** Used QwQ-32B-Preview to generate Thought/Action/Observation traces from MuSiQue, HotpotQA, and Natural Questions. Filtered trajectories based on final-answer correctness, yielding **about 20,000 examples**.
- **Supervised fine-tuning (SFT).** Fine-tuned GLM-4-9B on retrieval traces and general instruction data. Applied observation loss masking so that only reasoning steps and tool calls contributed to the trajectory loss.
- **Evaluation.** Built a long-context vs. RAG evaluation framework with contexts extended to 50k+ tokens. Evaluated ReaRAG across six QA benchmarks using LLM-as-a-Judge, achieving **48.5%** average accuracy, a 3.8 percentage-point improvement over the baseline.
- **Ablation & error analysis.** Conducted ablation experiments to validate training effectiveness (48.5% vs. 40.5% accuracy); analyzed retrieval depth and failure cases across reasoning and tool-use steps.

Selected Projects

Venture Buddy | Agent Harness for VC Research Jul – Oct 2026

Project lead | Domain-specific agent harness, expert knowledge modeling, and self-improving research strategy

- **Domain-specific agent harness.** Led development of the harness using the Claude Agent SDK. Coordinated specialist subagents with skills and hooks and connected them to research tools via MCP.
- **Expert-guided agent design.** Worked with venture investors to distill their sector research methods into reusable skills and commands for PEST analysis and evidence checks. Agents screened sectors for opportunities and produced reports that cited sources and identified areas to monitor.
- **Self-improving harness experiment.** Designed a Builder-Runner-Reviewer workflow that separated research execution from strategy updates. Tested an LLM judge on run logs and evidence tables. Quality judgments on open-ended research were subjective, so the scores provided too weak a feedback signal for automated updates. Shifted toward expert-guided skills refinement.

ASPICE Coding Agent | Automotive Unit Test Generation Mar – Apr 2026

Project lead | Pi-based multi-agent orchestration and execution feedback loops

- **Test coverage.** Led a coding agent project for an in-vehicle Kotlin module. Line coverage rose from **28.0% to 86.8%** and branch coverage from **22.3% to 71.3%**. The client signed off on phase one.
- **Multi-agent orchestration.** Built a reusable Pi extension coordinating Explorer, Writer, and Reviewer agents. Integrated CLI tools and skills for builds, tests, and coverage checks; isolated Writer and Reviewer contexts to reduce potential bias from shared context during test review.
- **Feedback loop.** Connected the agent to the client's build environment. Compiler errors, failing tests, and coverage reports guided the next edit, and the agent reran the tests to verify each fix.

AI Pet Video Generation | In-Vehicle Entertainment Sep – Dec 2025

Multimodal generation workflows, asynchronous task scheduling, and VLM quality assessment

- **Multimodal agent workflow.** Built an asynchronous image-to-video generation pipeline with multi-model routing, task scheduling, and state management using Redis, PostgreSQL, workers, and object storage. Integrated a vision-language model (VLM) for output quality assessment.

Publications

ReaRAG: Knowledge-guided Reasoning Enhances Factuality of Large Reasoning Models with Iterative Retrieval Augmented Generation
Jinxin Liu*, **Zhicheng Lee***, Shulin Cao, Jiajie Zhang, Weichuan Liu, Xiaoyin Che, Lei Hou, Juanzi Li
International Joint Conference on Neural Networks (IJCNN), 2026

AtomR: Atomic Operator-Empowered Large Language Models for Heterogeneous Knowledge Reasoning
Amy Xin, Jinxin Liu, Zijun Yao, **Zhicheng Lee**, Shulin Cao, Lei Hou, Juanzi Li
31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Vol. 2, 2025

DiaKoP: Dialogue-based Knowledge-oriented Programming for Neural-symbolic Knowledge Base Question Answering
Zhicheng Lee*, Zhidian Huang*, Zijun Yao, Jinxin Liu, Amy Xin, Lei Hou, Juanzi Li
Conference on Information and Knowledge Management (CIKM), Demo Track, 2024

Deep Semi-Supervised Learning via Dynamic Anchor Graph Embedding Learning
Zihao Wang, Enmei Tu, **Zhicheng Lee**
International Joint Conference on Neural Networks (IJCNN), 2021

* Equal contribution.